



Module
Presenter's Manual
for
**Building Full Stack
Applications**

***Effective from: March 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Explain Full Stack Development
- Describe setting up Mongoose for MongoDB Integration
- List and explain the technologies for Full Stack development
- Explain setting up back end server
- Build front end for Full Stack development
- Identify the communication between front end and back end
- Explain the implementation of file uploads within MERN Web applications
- Explain handling of authorization and authentication
- Explore error handling in React.js and Express.js
- Explain HTTP requests and responses
- Learn the deployment strategies for Full Stack applications
- Explain the testing strategies for Full Stack development

2. Information on Session Allocation

Module	Online (No. of Hours)	Self-study (No. of Hours)
FSTACK	40	8

Throughout this Presenter's Manual, the module **Building Full Stack Applications** will be referred to as **FSTACK**.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

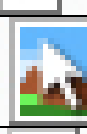
➤ Week-Wise Schedule

Week	Day 1		Day 2	Day 3	Day 4	
1	Session 1 FSTACK – TL1		Session 2 FSTACK – TL2	Session 3 FSTACK – TL3	Session 4 FSTACK – TL4	
2	Session 5 FSTACK – TL5		Session 6 FSTACK – TL6	Session 7 FSTACK – TL7	Session 8 FSTACK – TL8	
3	Session 9 FSTACK – TL9		Session 10 FSTACK – TL10	Session 11 FSTACK – TL11	Session 12 FSTACK – TL12	
4	Session 13 FSTACK – TL13	S1- S2	Session 14 FSTACK – TL14	Session 15 FSTACK – TL15	Session 16 FSTACK – TL16	
5	Session 17 FSTACK – TL17		Session 18 FSTACK – TL18	Session 19 FSTACK – TL19	Session 20 FSTACK – TL20	S3-S4

TL: Online Session

S: Self-Study Session (over and above the module duration)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	FSTACK – TL1	All the topics as listed below from Session 1 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 1 – Introduction to Full Stack Applications</u> ➤ Explain Full Stack development ➤ Describe ReactJS and its features ➤ Explain Node.js and its features ➤ Explain Express and its features ➤ Explain MongoDB and its features	<u>Building Full Stack Applications</u> SG – Session 1 XP – Session 1 TG – Session 1
2	FSTACK – TL2	All the topics as listed below from Session 2 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 2 – Setting Up the Development</u> ➤ Explain how to install and configure of MongoDB ➤ Describe how to set up Mongoose for MongoDB integration ➤ Explain how to install and configure ReactJS, Node.js, and Express	<u>Building Full Stack Applications</u> SG – Session 2 XP – Session 2 TG – Session 2
3	FSTACK – TL3	The Try It Yourself questions from Session 1 & 2 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 1 & 2
4	FSTACK – TL4	All the topics as listed below from Session 3 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 3 – Technologies for Full Stack</u> ➤ Explain animation, transition, and their implementation in ReactJS	<u>Building Full Stack Applications</u> SG – Session 3 XP – Session 3 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain styling using CSS and its implementation in MERN ➤ Define responsive design and media queries in ReactJS	
5	FSTACK – TL5	All the topics as listed below from Session 4 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 4 – Building Back End</u> ➤ Explain how to set up the back end server ➤ Define API endpoints for data retrieval and manipulation ➤ Explain how to connect to the database and handle data persistence	<u>Building Full Stack Applications</u> SG – Session 4 XP – Session 4 TG – Session 4
6	FSTACK – TL6	The Try It Yourself questions from Session 3 & 4 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 3 & 4
7	FSTACK – TL7	All the topics as listed below from Session 5 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 5 – Building the Front End</u> ➤ Explain how to create the project structure and file organization ➤ Describe the implementation of the user interface using HTML, CSS, and JavaScript ➤ Explain the utilization of frontend frameworks	<u>Building Full Stack Applications</u> SG – Session 5 XP – Session 5 TG – Session 5
8	FSTACK – TL8	All the topics as listed below from Session 6 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 6 – Integrating Front End and Back End</u> ➤ Identify the communication	<u>Building Full Stack Applications</u> SG – Session 6 XP – Session 6 TG – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		between front-end and back-end ➤ Describe the procedure to make API requests ➤ Explain handling of data responses	
9	FSTACK – TL9	The Try It Yourself questions from Session 5 & 6 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 5 & 6
10	FSTACK – TL10	All the topics as listed below from Session 7 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 7 – File Upload Concepts</u> ➤ Explain file upload concepts and requirements ➤ Explain how to handle file upload on the client-side using React.js ➤ Define file upload routes and middleware in Node/Express ➤ Explain how to validate file uploads and managing file storage ➤ Explain the handling of different file types and sizes	<u>Building Full Stack Applications</u> SG – Session 7 XP – Session 7 TG – Session 7
11	FSTACK – TL11	The Try It Yourself questions from Session 7 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 7
12	FSTACK – TL12	All the topics as listed below from Session 8 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 8 – Authentication and Authorization in React.js and Node/Express</u> ➤ Distinguish the concepts of authentication and authorization ➤ Describe implementation of user registration and login	<u>Building Full Stack Applications</u> SG – Session 8 XP – Session 8 TG – Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		functionality ➤ Describe how to handle password hashing ➤ Explain handling of authorization and authentication errors	
13	FSTACK – TL13	The Try It Yourself questions from Session 8 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 8
	S1-S2	All the Assignment topics from Session 1 to Session 8 of <i>Building Full Stack Applications</i> course should be covered in this session.	<u>Building Full Stack Applications (OnlineVarsity)</u> Session 1-8
14	FSTACK – TL14	All the topics as listed below from Session 9 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 9 – Handling Errors and Validation in React.js and Express.js</u> ➤ Describe Error handling in React.js and Express.js ➤ Explain validation techniques ➤ Explain how to list ways to log and report errors	<u>Building Full Stack Applications</u> SG – Session 9 XP – Session 9 TG – Session 9
15	FSTACK – TL15	The Try It Yourself questions from Session 9 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 9
16	FSTACK – TL16	All the topics as listed below from Session 10 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 10 – HTTP Requests and Responses</u> ➤ Explain different types of HTTP requests in React.js ➤ Describe how to handle request data and query parameters in Express.js	<u>Building Full Stack Applications</u> SG – Session 10 XP – Session 10 TG – Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain sending JSON responses from Express.js ➤ Explain pagination and filtering in API requests ➤ Describe how to handle API versioning and content negotiation	
17	FSTACK – TL17	The Try It Yourself questions from Session 10 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 10
18	FSTACK – TL18	All the topics as listed below from Session 11 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 11 – Deployment and Optimization</u> ➤ Explain deployment strategies for full stack applications ➤ Define performance optimization techniques (ReactJS and Express.js applications) ➤ Define caching strategies and CDN integration ➤ Define database performance tuning and indexing ➤ Describe monitoring and error tracking in production	<u>Building Full Stack Applications</u> SG – Session 11 XP – Session 11 TG – Session 11
19	FSTACK – TL19	All the topics as listed below from Session 12 of <i>Building Full Stack Applications</i> book should be covered in this session. <u>Session 12 – Testing in Full Stack Development</u> ➤ Explain unit testing for Full Stack applications ➤ Explain integration testing ➤ Describe end-to-end testing	<u>Building Full Stack Applications</u> SG – Session 12 XP – Session 12 TG – Session 12
20	FSTACK – TL20	The Try It Yourself questions from Session 11 and Session 12 of <i>Building Full Stack Applications</i> book should be covered in this session.	<u>Building Full Stack Applications</u> Session 11 & 12

Session No.	Session Title	Session Details	Deliverables' Mapping
	S3-S4	All the Assignment topics from Session 9 to Session 12 of <i>Building Full Stack Applications</i> course should be covered in this session.	<u>Building Full Stack Applications</u> (OnlineVarsity) Session 9-12

5. Library References

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| ➤ Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node by Vasan Subramanian |
| ➤ Beginning MERN Stack Development by Greg Lim |
| ➤ MERN Stack Programmer Notebook (Dark Mode): Javascript Coding Journal Achieve Your Goals Gift For Those Who Want to Learn MERN Stack (8.5 x 8.5) by MERN coders |

~~~ End of Document ~~~